Appnote 0025 C47 / R47 REGISTER EXPORT DATA FILE (.d47) FORMAT 2026-06-16

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This appnote briefly explains the .d47 data interchange file format.

The data file is written to the /DATA folder and the files are expected to be in that folder.

Each register is written as three lines:

1. its name (RX, R007, or a variable name)
2. its type (a 4-character code, optionally with an angle/polar tag)
3. its value (matrices add a "rows cols" line and then the element rows)

VALUE TYPES

Real - a plain number.

```
RA
Real
3.14159

R022
Real
6.02214076E23
```

Cplx - a complex number. We write it like (3-i4). Also accepted (all the same value): 3-i4 or (3 - i4) or 3 -4 and more whitespace variants. See below for exhaustive list.

```
RB
Cplx
(3-i4)

R023
Cplx
(1.6e-19-i2.5e-19)
```

LonI - a big whole number, as big as you like, up to 1000 digits.

```
RC
LonI
1234567890123456789012345678901234567890
```

ShoI - a short integer. Written as a signed hex magnitude (with a 0x prefix), then #, then the base in decimal. So the value 255 stored in base 10 is written +0xff#10. Also accepted: the old form +0xff 10. Example: the value 255, register base 10.

```
RD
ShoI
+0xff#10
```

Stri - text.

```
RE
```

Stri
Hello world

THMS - written as HMS coded decimal as HH.MMSSdddd. So 13:30:45.5 is written 13.30455.

RI
THMS
13.30455

DYMD - a date coded decimal DYMD (year.monthday), DDMY (day.monthyear), or DMDY (month.dayyear). All three are accepted. We export in the date mode of the calculator. Example, 15 June 2026 in YMD mode:

RJ
DYMD
2026.0615

Rema - a real matrix (or vector). A "rows cols" line, then the elements. Whitespace between elements does not matter: tabs, spaces, newlines, any mix. Elements can be vertical or horizontal, or in rows. The "rows cols" count decides where rows break. We write tab-separated rows for readability.

```
RX
Rema
2 3
1	2	3
4	5	6
```

Cxma - a complex matrix. Same layout, each cell a complex number. If free spacing is used, complex numbers must be in parenthesis. If one line per element is used, parenthesis are not required.

```
RY
Cxma
2 2
(1+i0)	(0-i1)
(0+i1)	(1+i0)
```

RXFN - a very long real (extended precision) used by the XFN functions. One decimal value with an E exponent, same angle tag as Real. Loads only to the stack (restores X, Y, Z), so the name must be RX. Export writes X automatically.

RX
RXFN:DEG
3.14159265358979323846...E+500

ANGLE / POLAR TAG (on Real, Cplx, and vectors / complex matrices)

When a number carries an angle mode, it is appended after a colon.
A trailing "p" means polar form.

```
:DEG     degrees
:RAD     radians
:GRAD    gradians
:DMS     degrees / minutes / seconds
:MULTPI  multiples of pi
```

Example - a real angle in degrees:

RZ
Real:DEG
90

Example - a complex matrix, polar, angles in degrees (the "p" = polar):

RT
Cxma:DEGp
1 2
(2+i30)   (5+i45)

A plain Real or Cplx with no colon means no special angle mode is attached.

REAL NUMBER INPUT

Real entry rules:

Exponent entry: E or e marks the power of ten: 6.02214076E23 means 6.02214076 times 10^23
No separators allowed: a number must be continuous, with no spaces and no digit-group separators.
Real, Cplx, Rema, Cxma, THMS, DYMD, DDMY, DMDY all use real numbers underneath, with the rules above.
The decimal mark may be a dot or a comma on input (3.14 or 3,14); exports always use a dot.

COMMENTS

A comment is the keyword "Cmnt" on its own line, followed by one line of text. The loader reads and drops both. Multiple consecutive comment lines allowed.
Comments go before a section or before a register's name line. Never inside an entry: not between an entry's name, type and value, and not among matrix rows. We do not write comments in exports. They are for hand notes.

Cmnt
Exported from the lab calculator, 16 June 2026
Cmnt
Stack snapshot before the regression run

Do not place a comment inside a NAMED_VARIABLES section: a Cmnt line there is taken as a variable name and will raise an error.

See example file below:

SPECIAL

Conf - a stored configuration. You cannot write this by hand and should never put it in a data file. If a Conf entry IS present, the loader does NOT ignore it: it decodes the data straight into the configuration, overwriting calculator settings. A stray or hand-edited Conf line can silently break your calculator setup. Leave Conf out of data files entirely.

???? - only appears in exports if something broke during a save. Never write it.

COMPLEX number format

Accepts (3-i4) | 3-i4 | +3+i4 | -3-i4 | (+3+i4) | (-3-i4) | -2.5e-3+i4 | (-2.5e-3-i4) | i4 | (i4) | (3 - i4) | (1.5e2-i2.5e-3) | 0+i0 | -7+i0 | (3 -i4) | 3 -i4 | +3 +i4 | -3 -i4 | (3 - i 4) | +3 + i4 | 3 - i 4 | (-2.5e-3 - i 4)

Also accepts the state file format space form (real and imaginary separated by a space, no i): 3 4 | 3 -4 | -2.5 1e3

(parentheses optional, leading sign optional)

(parenthesis required when matrix whitespace layout is used)

REGISTERS AND VARIABLES

The name line entry is either a numbered register, a lettered register, or a variable name.

Numbered registers: R followed by the number. We write three digits (R000 to R099), but on read the digits are free: R7, R07 and R007 all mean register 7.

Lettered registers (100 to 125): We write the short letter form RX RY RZ RT RA RB RC RD RL RI RJ RK RM RN RP RQ RR RS RE RF RG RH RO RU RV RW. The numeric form (R100 etc.) is also accepted on read. R100 is RX.

Variable names: saved and loaded by name (in a NAMED_VARIABLES section), so they land in the right variable on any calculator. A name is 1 to 7 characters. Do not place a comment inside a NAMED_VARIABLES section: a Cmnt line is taken as a variable name, not a comment and will raise an error.

COMMENTED FILE EXAMPLE

DATA_FILE_REVISION (this is fixed)
0 (this is fixed)
Cmnt (this is the main comment command)
MAIN COMMENTS GO here, BEFORE GLOBAL_REGISTERS: between the revision header and the first section
GLOBAL_REGISTERS (this is fixed)
2 (this is the number of registers that follow)
Cmnt (this is the optional register comment command and line)
REGISTER RELATED COMMENTS GO here, BEFORE the Rnnn command: between two registers
Cmnt (this is a second optional comment command and line)
This is another comment line
Cmnt (this is a another optional comment command and line)
This is one more comment line, always headed up with Cmnt, before the Rnn command
RY (this is the destination register, example RX, RM, RK, R020, R099)
Cplx (this is the command stating the next line is a complex input)
(3-i4)
RZ (this is the destination register, example RX, RM, RK, R020, R099)
Rema (this is the command stating the next line is a real matrix input)
2 2 (this is specific to matrix input, rows cols)

1 2 (these are the elements, matrix input is whitespace-insensitive, it can be 1 2 3 4 or 1 2 3 \n 4, or 1 \n 2 \n 3 \n 4 \n, etc.)

3 4

WORKED EXAMPLE 1

Export from C47/R47: example to create a 100-digit version of 3^-4:

CLRSTK 3 X->XFN 4 X->XFN XCHS X1/X XPOWER XRSD 100 EXPxfnx (Export XFN register X, enter file name A.d47)

File A.d47:

CODE: SELECT ALL

```
DATA_FILE_REVISION
0
GLOBAL_REGISTERS
1
RX
RXFN:DEG
1.234567901234567901234567901234567901234567901234567901234567901234567901234567901234567901234567901E-2
```

WORKED EXAMPLE 2

Export from C47/R47: is a lazy way of getting a complex matrix, just get the A-matrix from the ELEC menu. HOME ELEC [A]:

ELEC [A] EXPreg X (Export register X, enter file name B.d47)

File B.d47:

CODE: SELECT ALL

```
DATA_FILE_REVISION
0
GLOBAL_REGISTERS
1
RX
Cxma
3 3
(1+i0)  (1+i0)  (1+i0)
(1+i0)  (-0.5+i0.8660254037844386467637231707529362)  (-0.5-i0.8660254037844386467637231707529362)
(1+i0)  (-0.5-i0.8660254037844386467637231707529362)  (-0.5+i0.8660254037844386467637231707529362)
```

WORKED EXAMPLE 3

Export from C47/R47: is a factorisation result, this time of 45!:

45 x! FACTORS EXPreg X (Export register X, enter file name B.d47)

File C.d47

CODE: SELECT ALL

DATA_FILE_REVISION

0

GLOBAL_REGISTERS

1

RX

Rema

2 14

2 3 5 7 11 13 17 19 23 29 31 37 41 43

41 21 10 6 4 3 2 2 1 1 1 1 1 1

WORKED EXAMPLE 4

Manually populated (used internet sources and a template from above)

Load the file: IMPORTr (file PELL661.d47)

DATA_FILE_REVISION

0

Cmnt

Pell x^2 - 661 y^2 = 1, fundamental solution. Fermat (1657) posed these to English mathematicians.

Cmnt

Compute R001^2 - 661 * R002^2. R001 is 38 digits; result is exactly 1.

GLOBAL_REGISTERS

2

R001

LonI

16421658242965910275055840472270471049

R002

LonI

638728478116949861246791167518480580

RCL 01 x^2 661 RCL 2 x^2 -

And the result is 1!

USER PROBLEM 1:

Write files to import the CUBE42 problem solution: Add the cubes of the following three numbers:

Sum of three cubes for 42 (Booker & Sutherland, 2019; ~1e6 CPU-hours, Charity Engine).

Compute R001^3 + R002^3 + R003^3. Each cube is ~51 digits; they cancel to exactly 42.

-80538738812075974

80435758145817515

12602123297335631

USER PROBLEM 2:

Write files to import the CUBE3 problem solution: Add the cubes of the following three numbers:

Sum of three cubes for 3, third representation found 2019 (after only 1,1,1 and 4,4,-5).

Compute R001^3 + R002^3 + R003^3. Cubes are ~63 digits; cancel to exactly 3.

569936821221962380720

-569936821113563493509

-472715493453327032

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